

Soil & Air Environmental Telemetry Studio

Multi-Sensor Fusion Engine, VPD Analytics & Container Soil-Hydrology Physics

Experiment: Indoor Unplanted Soil Pot	Hardware: ESP32 Dual Probe Suite (ADC1)
Location: Near Window (Natural Airflow)	Telemetry URL: soil-air-dashboard.pages.dev
Direct Sun Timing: Incident After 1:30 PM	Author / Research: Apurba Maity & Ciel

1. Fault-Tolerant Dual-Sensor Fusion Architecture

Relying on a single capacitive or resistive sensor creates a fatal single point of failure: capacitive probes suffer from 555-timer oscillator lockup and cold-solder jitter, while resistive probes suffer from galvanic salt corrosion. The new **Dual-Sensor Fusion Engine** cross-validates both probes and dynamically re-weights confidence in real time:

1. Boundary & Plausibility Check	Capacitive ADC $\in [700, 2300]$ & Resistive ADC $\in [400, 3800]$. Out-of-bounds readings immediately drop confidence to 0.0 (detects open circuit or short).
2. Step-Jump Glitch Detection	If Capacitive jumps by $\Delta\theta > 25\%$ within 10 seconds while Resistive stays stable, the glitch detector shifts weights: $W_{cap} = 0.15$, $W_{res} = 0.85$.
3. Dynamic Consensus Weighting	Equation: $\theta_{fused} = (W_{cap} \times \theta_{cap} + W_{res} \times \theta_{res}) / (W_{cap} + W_{res})$ Normal Weights: $W_{cap} = 0.80$ (salinity immune), $W_{res} = 0.20$ (continuity anchor).
4. Thermal Dielectric Offset	Equation: $\theta_{final} = \theta_{fused} - 0.35 \times (T_{soil} - 25.0^{\circ}\text{C})$ Corrects for water permittivity reduction as soil warms ($\partial\epsilon/\partial T \approx -0.35\%/^{\circ}\text{C}$).

Why the Filtered Value is Always Lower Than Raw Percentage Readings

- Power-Law Calibration Curvature:** For normalized ratio $u \in (0, 1)$, $u^{1.15} < u$. A raw 70% reading becomes 66.3%, correcting fringe field distortion.
- Elevated Soil Temperature:** In your room, soil temperature is **29.5°C to 32.0°C**. The thermal compensation term $-0.35 \times (31.0 - 25.0)$ subtracts **2.1%** to eliminate false thermal capacitance.

2. Vapor Pressure Deficit (VPD) & Atmospheric Drying Demand

Vapor Pressure Deficit (VPD in kPa) measures the difference between saturated vapor pressure and actual atmospheric vapor pressure. It is the primary thermodynamic driver of soil surface evaporation.

VP_{sat} (Tetens Formula):	$VP_{sat} = 0.61078 \times \exp((17.27 \times T_{air}) / (T_{air} + 237.3))$ [kPa]
Actual VP (VP_{act}):	$VP_{act} = VP_{sat} \times (RH / 100)$ [kPa]
Vapor Pressure Deficit:	$VPD = VP_{sat} \times (1 - RH / 100)$ [kPa]

3. Unplanted Container Soil Physics: The 1:30 PM Direct Sun & Nocturnal Capillary Upflow

In your unplanted indoor container near the open window, soil moisture follows three coupled daily physical phases:

1. Morning Baseline (06:00 – 13:30)	Diffused room daylight, ambient window airflow. Air temp $\approx 30.5^{\circ}\text{C}$, RH $\approx 76\%$. Steady, gentle baseline surface drying ($d\theta/dt \approx -0.2\%/hr$).
2. Solar Pulse (13:30 – 17:30)	Direct afternoon sun hits pot surface. Soil temp climbs to 32°C – 39°C . Infrared radiation rapidly evaporates topsoil moisture ($d\theta/dt$ reaches -1.5% to -3.0%/hr , sharp amber valley).
3. Nocturnal Recharge (20:00 – 05:00)	No sun, air cools, window humidity reaches 80–84% RH. Topsoil was dried by afternoon sun (matric suction). Water from damp bottom of pot moves UPWARD by capillary action , while humid air adsorbs into pores ($d\theta/dt \approx +0.8\%$ to +1.5%/hr , emerald peak).

4. Research Lab Studio Graphs Reference

Soil Thermal Hysteresis (T_{soil} vs T_{air})	2D phase portrait loop. The enclosed loop area measures volumetric heat capacity of the water in the pot. Wet soil creates a wide loop due to water's high specific heat ($4.184\text{ J/g}^{\circ}\text{C}$); as soil dries, the loop collapses into a flat diagonal line.
Moisture Dynamics Rate ($d\theta/dt$ in %/hr)	5-point rolling central-difference derivative. Amber negative bars show afternoon solar evaporation; Emerald positive bars show nocturnal capillary recharge.
Dew Point Margin (ΔT_{dew})	Psychrometric distance to 100% saturation. When ΔT_{dew} approaches 1.5°C (red dashed line), ambient air cannot absorb more vapor, driving vapor adsorption into the soil.
Dielectric vs Conductive Discrepancy	Tracks $\theta_{\text{Cap}} - \theta_{\text{Res}}$ over time, monitoring dissolved salt ion concentrations and electrode passivation.

5. Gravimetric Scale Weight Cross-Validation Protocol

Evaporation Mass Loss Rate [g/hr] = $\Delta W / \Delta t = (W(t_1) - W(t_2)) / (t_2 - t_1)$

Telemetry Rate Proportionality: $d\theta/dt \approx -k \times (\Delta W / \Delta t)$

Logging total pot weight on a digital scale allows calibrating the scalar constant k so the dashboard can directly report **real-time water loss in grams/hour**.